

Housing Markets and Inequality: Identifying the Distributional Mechanisms of Urban Agglomeration

Abstract

Housing policy is widely assumed to reduce urban inequality, but the mechanism through which it does so, and whether it operates at all, remains unclear. Using a spatially explicit agent-based model calibrated to twenty-seven Brazilian metropolitan areas, we compute exact counterfactual changes in inequality and housing affordability across 1,596 simulations of housing-improvement and credit-rate policies. The average policy effect on inequality is negligible, but this null conceals two distinct and opposing channels: configurations that raise the rental burden raise inequality, while those that raise house prices lower it. The rental-affordability channel operates at every geographic level and is strongest in peripheral municipalities; the housing-wealth channel operates in the metropolitan area as a whole and in its urban core, but not in the periphery. The two policy levers examined, the regulated credit rate and the housing-improvement programme, reach the distribution through the affordability channel to a statistically indistinguishable degree. What determines the distributional reach of housing policy is therefore how it reshapes housing costs and asset values, not whether it is implemented.

Keywords: urban agglomeration, housing affordability, income inequality, agent-based model, spatial sorting

JEL: R21, R31, D63, C63

1. Introduction

Urban agglomeration is widely associated with productivity gains, economic growth and innovation [21; 37], yet its distributional consequences remain insufficiently integrated into theories and policy debates on agglomeration-driven development [28; 39]. Mainstream analysis tends to emphasise the trade-off between agglomeration-driven productivity and a more even distribution of resources and people across the territory [18; 38; 27]. This

gap is particularly evident at the urban scale. Most research on inequality has traditionally focused on the national level, although most people live in cities, where their experience of inequality is shaped by their immediate local environment [22]. Territorial inequality has thus become a comparatively neglected yet important source of the recent challenge to the political and economic system [37; 28].

Addressing these linkages requires attention to housing and living costs, which mediate how city-generated output is distributed across households and space, with direct implications for spatial justice, social equity, and productivity [39]. Housing plays a major role in inequality dynamics because it is simultaneously a capital asset linked to personal wealth, a consumption expenditure that constrains household budgets, and a location tied to amenities and social networks [26].

Despite the central role that housing markets play in distributing the gains of urban agglomeration, little is understood about the mechanisms through which housing policy interventions translate into distributional outcomes, or about how these mechanisms depend on the spatial structure of the metropolitan area, an indeterminacy consistent with cities' character as complex systems whose aggregate patterns emerge from decentralised interactions [7; 11; 27]. This paper addresses three connected questions. First, do housing policy interventions systematically reduce income inequality across Brazilian metropolitan areas? Second, through which housing market channel does this reduction operate, and specifically, what role does improved housing affordability play in transmitting policy effects to distributional outcomes? Third, do the effects concentrate in the urban core or in peripheral municipalities, and does this spatial pattern vary across cities of different sizes and structures?

We organise the analysis around three sequential mechanism tests that map onto these questions: an *aggregate effect test*, a *wealth channel test* (whether policy-induced asset appreciation compresses the distribution), and an *affordability channel test* (whether reducing the share of household income absorbed by housing costs drives the effect). A fourth question, orthogonal to the three channels, asks whether the aggregate policy effect scales with city population as a proxy for agglomeration intensity.

Our main finding is that the average policy effect on inequality is indistinguishable from zero, but this aggregate null conceals two clear and opposing mechanisms. Across all 1,330 treated observations, policy-induced changes in rental affordability, the share of household income absorbed by housing

costs, are positively and significantly associated with changes in inequality. This association is stronger in peripheral municipalities than in urban cores. Policy-induced changes in house prices work in the opposite direction: at the metropolitan and core levels, configurations that raise asset values compress the distribution of permanent income, on which the Gini coefficient is computed. The two levers, the regulated credit rate and the housing improvement programme, transmit through the affordability channel to a statistically indistinguishable degree, so neither can be said to dominate the other. These findings suggest that the distributional reach of housing policy depends less on whether it is implemented than on how it reshapes the affordability landscape across the income distribution and across metropolitan space. City population, the proxy for agglomeration intensity, bears no systematic relationship to policy-induced inequality change.

The remainder of the paper is organised as follows. Section 2 reviews the literature on housing markets and inequality, situating the wealth and affordability channels tested below within it. Section 3 describes the PolicySpace3 model, covering agent types, market interactions, and calibration across twenty-seven cities. Section 4 presents the simulation design, the counterfactual identification strategy, the geographic decomposition, and the affordability mechanism specification. Section 5 reports results for each of the four tests. Section 6 discusses limitations and open questions. Section 7 concludes.

2. Housing Policy and Inequalities in the urban context

Housing market dynamics are among the most important factors behind the growth of inequality within cities and regions [18]. Because housing supply is difficult to expand in the short run, house prices have risen considerably in economically successful cities. These housing market dynamics also interact with, and are reinforced by, broader processes of financial-system change, further fuelling inequality in both wealth and income.

Housing-wealth inequality often arises from the accumulation of asset value by owner-occupiers over time, a process that is reinforced whenever housing policy privileges ownership over other tenure forms [11; 26]. Christophers [11] argues that the emergence of significant housing-wealth inequality in countries such as Sweden and the UK is rooted in policies that have systematically favoured homeownership, fueling persistent house-price inflation and, in turn, widening the gap between those who own housing assets and

those who do not. Curbing the demand for ownership at practically any cost is, in this view, a precondition for narrowing housing-wealth inequality, since that demand is itself the source of the upward pressure on prices that drives wealth accumulation among owners.

In this sense, part of the literature concerned with inequality notes that policymakers need to address both housing costs and the land-use regulations linked to housing scarcity [26; 27]. Additionally, recent literature seeks to uncover the links between inequality and different housing policy tools, such as rent control [15], housing vouchers, reform of the mortgage interest deduction, public housing provision [26], upzoning and deregulation of housing markets [38], upgrading and regularisation of slum areas and social housing programmes [30]. Nonetheless, the dominant narratives about the emergence of housing-wealth inequalities and existing policy responses have focused primarily on the home ownership dimension. At the same time, another important tenure form, the rental one, has been increasingly neglected [11]. Specific aspects of both channels and how agglomerative forces interact with them will be explored in the next subsections.

2.1. The Wealth Channel

Policy-induced improvements in housing quality or in the ease of acquisition tend to raise asset values without compressing the broader income distribution. Subsidies and credit schemes that ease homeownership function as a form of demand lubrication that, given persistently inelastic supply, mainly sustains or inflates prices rather than expanding effective access [11]. Because house prices in many advanced economies have appreciated well above inflation for decades, the resulting wealth gains have accrued disproportionately to existing owners, concentrating housing wealth among the wealthiest deciles [11], and, over the long run, reproducing inequality across generations through inheritance [2].

This appreciation propagates, rather than compresses, inequality through the dual nature of housing as both a consumption good and a capital asset, an effect modelled in consumer theory as an endowment effect capturing the pure wealth gain from rising asset values [17]. Owner-occupiers can use accumulated equity as collateral and access better amenities, advantages unavailable to renters [26]. Equity extraction to finance current consumption further channels price appreciation into household leverage, with poorer households pressured to match richer households' spending and fuel debt accumulation

[9], while easier housing finance also encourages speculative participation among those already positioned to benefit [41].

The same logic extends to supply-side deregulation. Blanket upzoning, often proposed to resolve spatial inequality through trickle-down effects, mainly unleashes demand from high-income buyers and accelerates gentrification rather than reducing inequality, replacing lower-quality stock with higher-value development in already desirable areas [38]. Evidence is more mixed at the margin, where rising rents in high-productivity areas can nominally compress wage inequality, but the accompanying amenity improvements from higher-income in-migration tend to restore most of the welfare gap [14]. Taken together, this literature suggests the wealth channel operates less through redistribution of income than through the endogenous reinforcement of asset-based advantage [17; 11; 26; 38; 14].

2.2. The Affordability Channel

The affordability channel operates on the cost side of housing rather than on its asset value, and its distributional bite tends to be more direct and evenly felt. Housing is one of the largest items in household budgets, and its share has been growing over time, particularly for poorer households [6; 27; 26], reflecting how unevenly individuals' endowments are distributed across housing, labour and credit markets [26]. Groups that spend a larger income share on housing tend to be underrepresented in expensive cities, while skilled workers, who devote a smaller share, end up overrepresented there [14].

Because housing costs are deducted from income before other necessities can be met, two households with identical earnings can face very different levels of effective welfare depending on the share of income their housing absorbs [39]. A high housing-expenditure share limits resources available for non-housing consumption, saving, or human-capital investment, so that changes in rental costs translate into a comparatively direct income effect on the disposable income of renting households, most of whom hold no offsetting housing asset, an effect positively related to consumption inequality [26; 17]. On the credit side, the expansion of mortgage lending to riskier borrowers nominally broadened access to homeownership but also fuelled the price appreciation that undermined affordability, with rising financing costs disproportionately burdening middle-income households [9; 41].

The rental channel concerns the inequality between tenure forms itself, that is, the systematic disadvantage renters face relative to owner-occupiers in security, cost and long-term wealth-building. Christophers [11] argues

that closing the ownership-rental gap is central to reducing housing-wealth inequality, a view complemented by evidence from cities such as New York, Paris and London that an adequate, affordable rental stock requires substantial public subsidy rather than market provision alone [38]. Rental cost increases in many metropolitan areas fall disproportionately on lower-income households [38]. This channel has direct implications for spatial sorting. Households with wealth or income can minimise the costs of moving and secure high-amenity locations, whereas poorer households have little choice over where they end up, with housing affordability identified as a key driver of the selective migration of vulnerable households into less advantaged areas [3].

2.3. Agglomeration, Centrality, and the Spatial Dynamics of Housing Inequality

In the intrametropolitan dimension, agglomeration and centrality play a distinctive role in housing-wealth inequality, as housing preferences increasingly favour access to centrally-located urban amenities, transport, and employment [38], intensifying competition for inner-metropolitan locations, raising housing prices there, and constraining affordability in a process that can lead to gentrification. Income distributions and housing costs scale with city size in a predictable way, such that the largest cities disproportionately agglomerate the highest income earners and highest housing costs while showing comparatively lower concentrations of low- and middle-income residents and low-cost housing, pushing out lower- and middle-income earners and making affordability, diversity and socio-spatial justice central urban policy concerns [39]. This pattern is reversed in most small and medium-sized cities.

The structural asymmetry between centre and periphery relates to wage dynamics, amenity preferences, and differential exposure to housing-cost shocks. Intraregional inequality tends to be higher, and the welfare of unskilled workers lower, in regions that are less geographically central or have a smaller supply of developable land and housing, and this pattern is associated with a higher skill premium in these regions [29]. As competition for centrally located housing intensifies, distance from the urban core imposes a growing penalty on house prices, pushing less-skilled workers toward outer suburbs and leaving peripheral households structurally more exposed than core households to housing-cost fluctuations, given weaker labour-market access and thinner cushions against price shocks [38]. If the housing market of the largest cities is affordable only to high-income residents, and low-cost

dwelling grow sub-linearly with city size, low-income residents there become not merely relatively but absolutely disadvantaged [39].

Spatial sorting of workers by skill and education, a key feature of core-periphery models in the New Economic Geography tradition, also shapes these inequalities [14; 27]. Central areas' capacity to attract high-skill workers is linked to the local amenities they provide [13; 14], which are capitalised into land and house prices [27]. Higher-educated workers accordingly sort into cities offering higher wages and better amenities despite higher housing costs, while lower-skilled workers sort into more affordable, lower-amenity locations [13; 14].

This pattern is particularly pronounced in Brazilian metropolitan regions. In the São Paulo Metropolitan Region, low-income residents are overrepresented in peripheral areas, and combined with weak public transport infrastructure, this raises their commuting times and reduces access to job opportunities, a spatial mismatch reinforced by heavy dependence on radial transit networks [23]. The sorting of workers by skill concentrates high-paid workers in a small number of cities, while the resulting rise in housing prices deters low-skilled migration and exacerbates inequality across cities and regions through cumulative causation [27].

Read together, this literature suggests that the distributional consequences of housing policy hinge less on the existence of an intervention than on which of two channels it operates through: asset appreciation, which reinforces owner advantage, or cost-of-occupancy, which acts more directly and evenly across tenure groups. Few studies isolate these channels empirically in observational housing-policy data; Section 3 explains why the simulation strategy adopted here suits this identification problem. This gap is particularly salient in the Brazilian context: excessive rental burden is now the largest single component of the national housing deficit, a shift the country's own housing-policy diagnosis reads as evidence that the binding constraint is no longer housing stock but the relationship between housing cost and income.¹ Yet *Minha Casa Minha Vida*, Brazil's main housing policy instrument, remains designed principally around ownership transfer rather than rental affordability, a bias the same diagnosis identifies as one of the programme's structural

¹Fundação João Pinheiro deficit accounting (61.3% of the national housing deficit attributed to excessive urban rental burden) and diagnosis prepared for the PlanHab 2025–2040 revision (Rodrigues, Cardoso, Krause et al., unpublished policy report).

bottlenecks. The analysis that follows exploits PolicySpace3’s exact counterfactual environment to test how each of the two channels identified above carries the policy’s distributional effect in this setting.

3. Model description: The PolicySpace3

The distributional consequences of housing policy are difficult to identify in observational data: cities that adopt interventions differ systematically from those that do not, policy timing correlates with local economic conditions, and the relevant counterfactual trajectory is never observed. An agent-based simulation resolves these challenges by construction [16; 12]. As the model generates complete trajectories under both treated and baseline configurations for every city and stochastic seed, exact counterfactual differences are observed rather than estimated, and no selection assumptions are required. The cost of this approach is that causal claims are model-internal, they hold to the extent that the simulated mechanisms approximate real-world processes, a condition addressed through multi-city calibration and empirical validation, described in Section 3.3 below.

PolicySpace3 (PS3) is a spatially explicit, open source agent-based model of metropolitan economies, designed to analyse the distributional consequences of housing, fiscal and other policies across Brazilian cities [36; 1]. It is designed to reflect empirical regularities and distribution patterns while allowing interactions among agents to occur within space and be mediated by it.

3.1. Agents and Heterogeneity

The model represents heterogeneous households and firms whose characteristics are drawn directly from empirical distributions, such as the Census [24] and the *Relatório Anual de Informações Sociais* [8], an administrative census of formal labour, transformed into distributions of population characteristics (age, location) and firm characteristics (sector, location), grounding the model’s inequality dynamics in observed socioeconomic structure.

Agents are generated with profiles drawn from empirical data and behavioural rules derived from theoretical models and heuristic definitions [16; 12]. Individuals provide labour and belong to households, which decide on consumption, savings, and housing; firms hire labour, use inputs from other firms via a regionalised input-output matrix [25; 34; 31], sell to firms and

consumers, and pay wages. The government (one regionalised agent per municipality) collects taxes and implements all policy features; the bank lends to households to finance home purchases, with credit directed exclusively at households and no firm-level borrowing.

3.2. Spatial Structure and Markets

Agents interact across goods, labour, housing, credit, and intermediate markets within a neighbourhood-level spatial grid, where residential sorting, production concentration, and housing prices (including explicit dwelling quality) emerge endogenously from market interactions [20] and government investment.

The five markets are interdependent. Wages set in the labour market determine household income, which governs both credit eligibility in the housing market and demand volumes in the goods market; housing costs feed back into disposable income and saving rates; and intermediate goods flows link regional economies through the input-output structure. The policy scenarios examined here enter this circular flow at two points (the regulated credit rate and the housing improvement programme) and propagate to affordability and inequality through the linkages identified in Section 5.

In the goods market, consumers browse a random subset of firms each period and buy from the closest or cheapest one, creating differentiated demand patterns in denser locations while retaining price competition.

In the labour market, firms hire through an iterative matching process where each candidate-firm pair is scored via a Cobb-Douglas function of qualification, commuting cost, and wage [35; 33]. The highest-scoring pairs are matched first [32], favouring denser population patterns closer to important, higher-paying firms.

The credit market channels the housing policies analysed here. The bank manages two funding sources: retail credit, using an exogenous interest rate (observed SELIC, forecast for future months) with perfectly elastic supply, and policy funds, limited to a calibrated percentage of municipality GDP (Section 4).

In the housing market, families search for housing, competing in sub-markets defined by income level [5]. House prices adjust endogenously based on location quality, vacancy duration, neighbourhood wealth, and aggregate vacancy conditions. Households decide to buy and stop renting based on savings and credit access (where the policy helps); transactions are bilateral and follow a negotiation process, described fully in [1].

Renting is the default tenure for families that have not bought a home, and the channel through which the affordability mechanism examined in this paper (Section 4.3) operates. Rent is a fixed proportion of the dwelling’s current sale price, so movements in the sale market pass directly through to rental costs. Prospective tenants are matched monthly against a sampled pool of vacant units, prioritised by permanent income; when nothing in the sample is affordable, the asking price is discounted rather than leaving the family unhoused. A settled family is prevented from moving into a costlier rental if doing so would push monthly rent above a fixed share of its permanent income, the affordability constraint whose policy-induced tightening or loosening is the mechanism this paper tests. Tenants draw first on cash, then savings, then bank deposits to cover rent; shortfalls register as default, and landlords receive rent net of a labour tax.

3.3. Calibration, Validation, and Multi-city Scope

The model is calibrated simultaneously for the Brazilian state-capital metropolitan areas using census microdata, firm-level location data, and regional input–output tables, so that cross-city differences in outcomes reflect differences in urban structure rather than model specification; the twenty-seven areas analysed here (Section 4) all share the resulting common parameter set. Validation follows [40] and [19]: (i) calibrate parameters and intervals using prior knowledge and data, (ii) validate outputs against empirically observed stylised facts, and (iii) retain the configuration that most consistently reproduces them. The target stylised facts are the negative relationship between prices, unemployment and output for macroeconomic coherence (Phillips Curve and Okun’s Law), and a set of housing-market indicators compared against empirically observed values.

Results show that the model is able to qualitatively replicate these regularities given the calibrated parameter set. Figure 1 sets simulated inflation and GDP growth against their empirical counterparts over 2012–2024, each z-scored to its own mean and standard deviation. The model is initialised in 2010 and is not driven by the realised shocks of the period, so the comparison is one of amplitude rather than of timing and no month-to-month correspondence is claimed; what it shows is that the simulated series fluctuate on a scale comparable to the observed ones rather than drifting or collapsing. The stylised-fact tests below carry the validation. The Phillips Curve slope is negative and statistically significant in every scenario, and the Okun coefficient is likewise negative and significant throughout; it is strongest under low

regulated credit rates (consistent with a more credit-responsive production environment) and weakest under high ones.

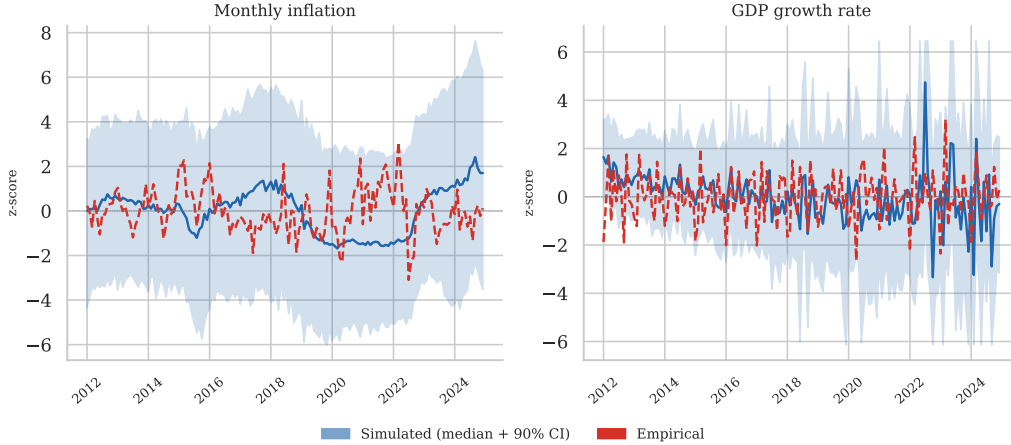


Figure 1: Normalised time-series validation. Simulated median and 90% confidence interval (blue band) versus the empirical series (dashed red), each z-scored to its own mean and standard deviation over 2012–2024; because the two sides are normalised independently, the width of the simulated band is not comparable to the amplitude of the empirical line. Left: monthly inflation rate. Right: monthly GDP growth rate. Empirical series from Ipeadata: the IPCA for inflation, and the month-on-month growth of monthly GDP at current prices, deflated by the IPCA so that both sides are in real terms, for growth; the latter is not seasonally adjusted.

To demonstrate this statistical adherence directly, Table 1 reports the Phillips Curve and Okun’s Law regressions in both pooled and baseline-scenario specifications. Both relationships are also stable when re-estimated separately within each of the six policy scenarios (Table A.5, Appendix Appendix A).

For the housing market, Table 2 compares seven simulated moments against empirical benchmarks. Three of the seven indicators fall within the empirical range; the four flagged indicators (the price-to-annual-income ratio, consumption-to-GDP, housing production per thousand inhabitants, and housing stock relative to permanent income) are the least adherent ones.

Some limitations in calibration are worth noting: the housing-production gap identified above recurs in the discussion of the paper’s substantive limitations in Section 6.1, while the consumption-to-GDP, housing-stock-to-income and price-to-income gaps are calibration-only misses that do not bear directly

Table 1: Phillips curve and Okun’s law: baseline specifications

Specification	$\hat{\beta}$	p -value
<i>Phillips curve</i> (prices on unemployment)		
Full panel, contemporaneous	-0.0012***	< 0.001
Full panel, city + scenario FE	-0.0012***	< 0.001
Baseline scenario, full horizon, contemporaneous	-0.0012***	< 0.001
Baseline scenario, full horizon, lagged	-0.0014***	< 0.001
<i>Okun’s law</i> (unemployment on output)		
Full panel	-0.203**	0.0103
Full panel, city + scenario FE	-0.203**	0.0103
Baseline scenario, full horizon	-0.191***	0.0089

Notes: panel regressions on monthly, city-level simulated series with metropolitan-area fixed effects and standard errors clustered by metropolitan area. The Phillips specification regresses monthly inflation on the unemployment rate; the Okun specification regresses monthly GDP growth on the month-on-month change in unemployment, first-differenced within each simulation so no difference is taken across runs. “Full panel” pools all six policy configurations, “baseline scenario” restricts to the reference configuration (medium interest rate, improvement policy inactive). “City + scenario FE” adds fixed effects for the interest-rate $\times$ improvement-policy cell; “lagged” uses X_{t-1} in place of X_t .

The first 18 months of every run are dropped as burn-in: unemployment falls steeply from its initialisation value over that transient, and including it dominates the estimated relationships.

Both relationships are negative and significant in every specification, consistent with the target stylised facts described in Section 3.3.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Housing market validation: simulated moments compared to empirical benchmarks. Simulated values report the median and the 10th–90th percentile range across all simulation runs. † indicates that the simulated range falls outside the empirical benchmark.

Indicator	Empirical range	Simulated median (p10–p90)	Source
Housing stock / GDP	1.5–2.5	1.87 (1.45–2.37)	^a
Price-to-monthly-wage ratio	60–150	91 (77–141)	^b
Price-to-annual-income ratio †	6.0–12.0	4.9 (4.1–5.9)	^c
Vacancy rate	0.08–0.13	0.09 (0.08–0.10)	^d
Consumption / GDP †	0.55–0.65	0.42 (0.32–0.47)	^e
Housing production (per 1,000 hab.) †	2.0–4.0	11.0 (7.7–12.7)	^f
Housing stock / permanent income †	5.0–8.0	3.9 (3.4–4.7)	^g

Notes: ^a Banco Central do Brasil, Relatório de Estabilidade Financeira (2019). ^b Fundação João Pinheiro, Déficit Habitacional no Brasil (2022). ^c Banco Central do Brasil, Relatório de Estabilidade Financeira (2023). ^d IBGE, Censo Demográfico 2010 and 2022. ^e IBGE, Contas Nacionais (2010–2024 average). ^f Câmara Brasileira da Indústria da Construção – CBIC (2019). ^g Authors’ estimate based on national household survey data.

on the mechanism this paper tests, since none of them enters the affordability ratio ΔM on which Equation 3 is estimated. The consumption-to-GDP entry sets mean monthly household spending, scaled by the number of households, against value added; the model’s expenditure side comprises household and government consumption, with no gross fixed capital formation and no external sector. The simulated macroeconomic series reproduce the qualitative regularities targeted in calibration, but match neither the level nor the dispersion of their empirical counterparts.

4. Simulation design and policy scenarios

The analysis draws on PolicySpace3 (PS3), described in Section 3. All twenty-seven of Brazil’s state-capital metropolitan areas (Áreas de Concentração Urbana, ACUs) are run with a common parameter set, so that cross-city differences in simulated outcomes reflect differences in urban structure and initial conditions rather than city-specific calibration choices.

The simulation horizon runs from 2010 to 2039, matching the first month of the empirical input data; the nearly three-decade span mirrors the multi-decade horizon typical of Brazilian state housing planning, exemplified by the national housing plan (PlanHab), originally designed for 2009–2023 and since revised toward 2040.

For each city and policy configuration, ten independent replications are run with different random seeds, six in the case of São Paulo, yielding a total of 1,596 simulations ($26 \times 6 \times 10$ plus 6×6). The stochastic replications serve as within-cell noise estimates and provide the seed-level variation used in the mechanism regression. Ten replications are used because the seed-to-seed standard deviation of the tail-window aggregates used below is larger than the policy effects themselves, so fewer seeds would leave the per-city and subsample estimates reported in Section 5 unresolvable. Section 5.3 shows that some of them are not resolvable even at ten.

The housing policy environment consists of three components. The Minha Casa Minha Vida (MCMV) programme, Brazil’s main federal social housing instrument [10], is held active and constant across all runs. It has been continuously in effect nationwide since 2009 and forms part of the baseline environment rather than a policy lever under test; this paper does not evaluate MCMV itself. Each month the model draws on a municipal fund financed from the federal budget (OGU), sized as a historically observed share of municipality GDP (25 % of GDP/12 for 2010–2016, 4 % for 2017–2022, and 9 %

for 2023–2025), which purchases the cheapest available newly built houses and transfers them free of charge to eligible renter households (bottom 38th percentile of permanent income, owning no housing) until the municipality’s new housing stock or monthly fund is exhausted. Eligibility is means-tested, but allocation among the eligible is not further ranked by income in the configuration used here, so within the eligible pool the order in which households are served carries no income gradient. Two further policy dimensions are varied experimentally across the six configurations:

1. **Housing improvement policy:** a binary switch activating a programme targeting households in the bottom 38th income percentile who occupy inadequate housing (quality index 0.5), the tier corresponding to Brazil’s empirical housing-inadequacy classification of structural, sanitary, or overcrowding deficiencies [4]. The 38th-percentile threshold mirrors MCMV *Faixa 1* eligibility, so the programme targets the same income segment as the acquisition programme above rather than an independently chosen cutoff. When active, it upgrades each eligible household’s dwelling to adequate condition (quality index 1.0) at a cost equal to the full market value of the works, financed by OGU transfers scaled to municipality GDP. The municipality contracts a local construction firm, which is paid for the works and consumes the same construction capacity that a new dwelling would, so the programme is a demand injection into the construction sector as well as a transfer of housing quality to the beneficiary. The works are the quality increment from 0.5 to 1.0 on the dwelling’s existing floor area and location, which under the model’s own price function is worth the dwelling’s pre-upgrade market price; paying that in full makes the contract capacity-neutral for the builder, who earns the same revenue per unit of capacity as it would by building a new dwelling and selling it. A dwelling is upgraded in a given month only when both the funds and a builder with the spare capacity to complete the work are available; households that miss out remain eligible in the following month. Because the upgrade raises the dwelling’s market value, the policy has a countervailing effect: improved stock may appreciate in price, partially offsetting the welfare gain for non-beneficiary tenants and buyers. The two values are *active* and *inactive*.
2. **Regulated housing credit rate:** the interest rate applied to SBPE (Sistema Brasileiro de Poupança e Empréstimo/Brazilian Savings and

Loan System) and FGTS (Fundo de Garantia do Tempo de Serviço/ Employment Severance Guarantee Fund) directed housing loans. These regulated rates are set independently of the basic rate *SELIC* and govern the financing cost of the main credit instruments available to lower- and middle-income homebuyers. All three scenarios follow identical historical series before 2020 and diverge thereafter, holding the following annualised rates:

- *low*: SBPE 4.0 % p.a., FGTS 3.0 % p.a.
- *medium* (reference): SBPE 5.0 % p.a., FGTS 4.0 % p.a.
- *high*: SBPE 6.0 % p.a., FGTS 5.0 % p.a.

The full factorial of these two dimensions produces the six configurations per city. The reference baseline is *inactive improvement policy* $\times$ *medium credit rate*, in which MCMV operates but no upgrade programme is active and credit follows the intermediate rate path; SBPE and FGTS lending volumes are held at their historical trajectory across all configurations, so differences between credit-rate scenarios reflect financing cost alone, not fund supply. The regulated rate enters the housing market as a purchasing-power screen rather than as an approval gate. Each month, every house-seeking family’s maximum loan is computed at the rate its income band qualifies for, and the sum of that loan and the family’s savings and deposits defines the set of dwellings it can consider at all; the bank’s own approval step then re-checks a constraint the choice set has already imposed. A lower rate therefore does not admit a marginal borrower who would otherwise be refused so much as it enlarges the choice set of every prospective buyer, generating variation in affordability that is independent of the improvement policy and that compounds or attenuates the improvement policy’s effects when both levers are active.

4.1. Counterfactual identification

The simulation framework produces complete time series of all endogenous variables for every combination of city, policy configuration, and random seed. This allows policy effects to be defined as exact counterfactual differences rather than estimated from cross-sectional or panel variation in observational data.

For each metropolitan area i , policy configuration p , month t and random seed s , the policy effect on any variable Z is defined as:

$$\Delta Z_{ipts} = Z_{ipts} - Z_{its}^0 \quad (1)$$

where Z_{its}^0 denotes the reference baseline trajectory (housing improvement inactive, intermediate credit rate) for the same city and seed. The five non-baseline configurations each yield a distinct ΔZ_{ipts} , spanning a gradient from purely credit-cost effects to the combined subsidy-plus-credit package. Because all trajectories share the same city population, initial conditions, and random seed, the only source of difference between a treated run and the baseline is the policy configuration; no selection assumptions are required and there is no risk of post-treatment bias. The three credit-rate scenarios follow identical historical rate series until 2020, so replications that share a seed must produce numerically identical trajectories up to that point; across the whole batch the maximum relative discrepancy between arms before January 2020 is zero to machine precision.

To characterise long-run distributional effects, monthly differences are averaged over the final five years of the simulation horizon:

$$\Delta Z_{ips} = \frac{1}{60} \sum_{t=T-59}^T \Delta Z_{ipts} \quad (2)$$

where T denotes the final month of the simulation. This window captures the policy’s long-run effect once transitional dynamics have largely settled, and corresponds to January 2035 – December 2039. Because the baseline configuration provides the reference trajectory and is excluded from the regression, the analysis uses $N = 1,330$ treated observations (five non-baseline configurations for each city–seed pair) at the metropolitan and capital levels, and $N = 1,080$ at the periphery level, where five single-municipality ACUs have no peripheral municipalities.

Figure 2 provides the empirical basis for this choice. Before policy divergence in 2020, ΔY fluctuates within a narrow band around zero (mean = -0.0002 , std = 0.0004), confirming that the counterfactual differencing eliminates pre-treatment trends. After 2020 the pooled mean effect drifts mildly negative, reaching annual means of about -0.0009 over 2026–2027, then reverses sign and oscillates within roughly ± 0.0007 for the remainder of the horizon; the window mean is $+0.0002$. The trajectory does not settle onto a stable long-run level, so the 60-month tail window is not a steady state. Section 5.4.1 re-estimates Equation 3 on two earlier windows to es-

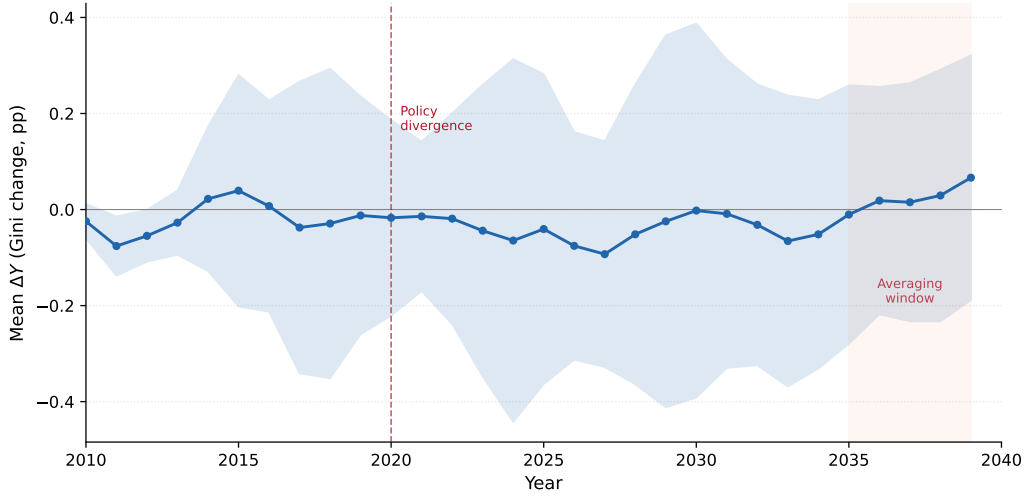


Figure 2: Trajectory of mean ΔY (policy-induced Gini change) over time, averaged across all seeds and metropolitan areas. The dashed vertical line marks 2020, when policy configurations begin to diverge. The shaded band marks the 60-month averaging window (2035–2039). Pre-2020 values fluctuate around zero (mean = -0.0002 , std = 0.0004), confirming baseline equivalence. The mean effect turns mildly negative after divergence, is most negative over 2026–2027, then reverses sign, leaving a window mean of $+0.0002$.

tablish how far the mechanism coefficient depends on where the window is placed.

4.2. *Geographic decomposition*

Each metropolitan area in the sample comprises one or more municipalities. To examine where within the urban fabric distributional effects concentrate, the analysis distinguishes three geographic levels: the full metropolitan area, the capital municipality (the primary urban core), and the periphery (all non-capital municipalities in the metropolitan area, aggregated with population weights).

Policy-induced changes in inequality and affordability are computed separately at each level. The capital-periphery gap additionally indicates the policy’s effect on spatial sorting: if the intervention disproportionately benefits lower-income peripheral households, it compresses the spatial income gradient; if effects concentrate in the core, peripheral inequality may remain unaddressed. Because this gap is constructed from counterfactual differences rather than observed levels, it avoids the confounding that complicates analogous comparisons in observational data.

4.3. *The housing affordability channel*

To identify the mechanism through which housing policy translates into distributional change, we relate the policy-induced change in the Gini coefficient to the policy-induced change in housing affordability:

$$\Delta Y_{ips} = \alpha + \beta \Delta M_{ips} + \mu_i + \varepsilon_{ips} \quad (3)$$

where ΔY_{ips} is the policy-induced change in the Gini coefficient for metropolitan area i , policy configuration p , and seed s ; ΔM_{ips} is the corresponding policy-induced change in the housing affordability ratio (the share of household income absorbed by housing costs); and μ_i are metropolitan-area fixed effects that absorb persistent structural differences across cities in size, income level, and urban form.

Housing affordability is selected as the mediating mechanism for three reasons, consistent with the affordability-channel literature reviewed in Section 2, which identifies the cost burden of housing, rather than asset appreciation, as the more direct determinant of household welfare [26; 39]. First, it is the direct target of the simulated interventions, which subsidise housing acquisition and improvement to reduce the cost burden of housing relative

to income. Second, affordability integrates price and income effects in a single quantity, avoiding multicollinearity between house values and household income entered separately. Third, the affordability ratio varies both across metropolitan areas and between core and peripheral municipalities, providing identifying variation at both levels of the analysis.

The coefficient β is identified from variation in ΔM_{ips} across policy configurations and stochastic seeds within each city, after removing city-level averages through the fixed effects. Two independent design dimensions generate this within-city variation: the housing improvement switch and the regulated credit rate, which directly enters the loan affordability gate and shifts the affordability ratio independently of the improvement policy. Because these dimensions are crossed by design, they provide orthogonal sources of variation and strengthen identification of β , which answers the paper’s central empirical question: conditional on urban structure, does a larger policy-induced improvement in housing affordability translate into a larger reduction in inequality?

Equation 3 is estimated separately for the full metropolitan area, the capital municipality, and the periphery, allowing the strength and direction of the affordability–inequality link to differ across geographic levels. This produces three estimates of β that together characterise how the spatial organisation of the city shapes the policy transmission mechanism.

4.4. Cross-city heterogeneity

The twenty-seven metropolitan areas span a wide range of population sizes, urban structures, and regional contexts, from large, dense cities such as São Paulo and Rio de Janeiro to smaller capitals such as Macapá and Palmas. The distribution of ΔY across cities is itself a primary result: strong dispersion would indicate that local urban structure shapes the distributional reach of national housing programmes, while convergence would suggest the mechanism operates similarly regardless of context.

A complementary analysis examines whether the magnitude and sign of ΔY correlate with city population, a proxy for agglomeration intensity along which Section 2 reviews evidence that income and housing-cost concentration scale predictably with city size [39; 27]. This association is reported graphically (Figure A.6) rather than as a formal regression, keeping the presentation focused on the mechanism in Equation 3; Section 6.5 discusses a density-covariate extension left for future work.

5. Results

5.1. *The affordability and wealth channels*

The average policy-induced change in the Gini coefficient across all 1,330 treated observations is +0.0002 (std = 0.011), indistinguishable from zero. Forty-nine percent of observations show a reduction in inequality; fifty-one percent show an increase. At the level of the aggregate mean effect, the housing policies examined here are distributionally neutral. This aggregate null, however, masks two mechanisms of opposite sign: cities where policy raises the rental burden experience larger increases in inequality, while cities where policy raises house prices experience reductions in it.

Policy configurations that increase the median rental burden relative to baseline are associated with higher inequality, while those that ease it are associated with lower inequality, with metropolitan-area fixed effects absorbing persistent cross-city differences so the relationship is identified from within-city variation across configurations and seeds. Figure A.3 in the Appendix plots the underlying observations.

Table 3 reports the full regression results at three geographic levels. The affordability coefficient is positive and highly significant throughout: $\hat{\beta} = +0.35$ for the full metropolitan area, +0.36 for the capital municipality, and +0.55 for the periphery (all $p < 0.001$). The stronger peripheral effect indicates that rental affordability changes have a larger distributional bite outside the urban core, consistent with lower-income peripheral households being more exposed to housing cost shocks. Explanatory power is modest and near-identical across the three levels (R^2 between 0.07 and 0.08): the channel is a reliable conditional relationship, not a description of most of the variance in ΔY , the bulk of which is seed-level stochastic variation within cells.

The stronger peripheral coefficient does not, however, translate into a directional difference in outcomes. Across the twenty-two multi-municipality areas the difference between peripheral and capital ΔY averages -0.0005 , with the periphery ending up worse off in nine of them, and a paired test does not reject equality ($p = 0.44$). The transmission is more elastic outside the core, as the spatial-sorting literature of Section 2 would predict; it does not follow that the policy leaves peripheral inequality systematically higher, and on these runs it does not.

House prices were also tested as an alternative mechanism variable representing a wealth channel, in which policy raises asset values and thereby

moves the distribution of permanent income. This specification, Panel A of Table 3, yields a negative and significant coefficient at the metropolitan and capital levels ($\hat{\beta} = -1.96 \times 10^{-4}$, $p < 0.001$, and -1.80×10^{-4} , $p < 0.001$), and a negative but insignificant one in the periphery (-1.44×10^{-4} , $p = 0.11$). Policy configurations that raise house prices therefore compress the simulated income distribution rather than widening it, and they do so in the urban core rather than outside it. The two channels are consequently not competing accounts of a single pathway: they are separate transmissions of the same policy shock, operating with opposite signs, in different parts of the metropolitan area, and on different populations. Section 6.3 discusses why the model produces a compressing wealth channel, which is not the sign the literature reviewed in Section 2 would lead one to expect.

Table 3: Housing wealth and affordability channels: regression results (Equation 3)

	ACP	Capital	Periphery
<i>Panel A: Primary mechanism</i> ($\Delta M =$ house price change)			
$\hat{\beta} \times 10^4$	-1.959*** (0.538)	-1.796*** (0.512)	-1.444 (0.898)
R^2	0.053	0.047	0.048
N	1330	1330	1080
Cities	27	27	22
<i>Panel B: Secondary check</i> ($\Delta M =$ median rent/income ratio, renters)			
$\hat{\beta}$	+0.3532*** (0.0651)	+0.3554*** (0.0645)	+0.5453*** (0.1024)
R^2	0.074	0.076	0.083
N	1330	1330	1080

Notes: OLS with metropolitan-area fixed effects (HC3 standard errors in parentheses).

ΔM and ΔY are 60-month window averages (2035–2039) of policy-induced changes (Eq. 1–2).

Panel A coefficients and standard errors are multiplied by 10^4 for legibility; ΔM there is a house price in model units, so the coefficient is small by construction. Periphery excludes five single-municipality ACPs (Boa Vista, Campo Grande, Manaus, Palmas, Rio Branco).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2. Interest rate versus housing improvement

To ask whether one policy lever drives the affordability channel more than the other, we estimate Equation 3 separately on two subsamples: observations where only the regulated interest rate varies (improvement policy inactive, $N = 532$) and observations where the improvement policy is active alongside interest rate variation ($N = 798$).

The interest-rate-only subsample yields $\hat{\beta} = +0.31$ ($p = 0.007$) and the improvement-active subsample $\hat{\beta} = +0.37$ ($p < 0.001$). Both are positive and significant, and they are not distinguishable from one another: a Wald test on the difference gives $z = -0.39$ ($p = 0.70$). The two levers transmit to inequality through the affordability channel at effectively the same rate, and this analysis provides no basis for ranking them. Section 6.4 discusses why the reach of the two instruments, which differ sharply in the breadth of the population they touch directly, nevertheless converges in this model.

5.3. Cross-city heterogeneity

Across the twenty-seven metropolitan areas, policy-induced Gini changes are heterogeneous: sixteen cities show mean inequality reductions and eleven show increases (Figure A.4, Appendix). Effects are individually statistically significant in only six of them (Curitiba, Natal, Palmas, Salvador, Teresina and Vitória) even at ten replications per cell.

That count is the reason this section reports dispersion rather than an ordering. The seed-level standard deviation of ΔY within a city-configuration cell has a median of 0.008, more than three times the standard deviation of the city means themselves (0.002): the stochastic variation the design is meant to average over is larger than the cross-city signal it would have to resolve. With twenty-one of twenty-seven cities indistinguishable from zero, the ranking of cities by mean ΔY is not identified by these data.

What the dispersion does establish is that the aggregate null of Section 5.1 is a composition effect and not a uniform absence of response. The correlation between metropolitan population (a proxy for agglomeration intensity) and mean ΔY is weak ($r = +0.05$), offering no support for a systematic gradient in policy effectiveness with agglomeration intensity. This null contrasts with evidence that income and housing-cost levels scale systematically with city size [39; 27]: the urban-scaling relationship documented in Section 2 governs the structure of the housing market, not, in this setting, the marginal effect of a policy shock on it.

5.4. Validation and robustness check

5.4.1. Window sensitivity

Because the average policy effect is small and the mechanism coefficient is estimated on a single 60-month averaging window, a natural concern is whether $\hat{\beta}$ is an artefact of that window rather than a stable model feature. We re-estimate Equation 3 (ACU level; affordability specification) on two earlier windows drawn from the interior of the post-divergence period shown in Figure 2: 2027–2031 and a narrower 2028–2030.

Table 4: Affordability-channel coefficient across averaging windows

Window	mean ΔY	$\hat{\beta}$	R^2	N
Tail (2035–2039, headline)	+0.0002	+0.3532*** (0.0651)	0.074	1330
Alternative (2027–2031)	-0.0002	+0.4550*** (0.0720)	0.102	1330
Alternative (2028–2030, narrow)	-0.0000	+0.4350*** (0.0730)	0.111	1330

Notes: Equation (3) (ACP level, secondary spec), re-estimated with the averaging window moved away from the long-run tail to the two earlier windows examined in Figure 2. HC3 SE in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4 reports the results: the affordability coefficient keeps its sign, its significance and its order of magnitude wherever the window is placed, at +0.35 in the tail window, +0.46 in the wider earlier window and +0.44 in the narrow one, each significant at $p < 0.001$. Its magnitude is not constant. The two earlier windows sit about a third above the tail estimate, a gap of roughly one and a half standard errors, and explanatory power rises with them ($R^2 = 0.074, 0.102, 0.111$). That gap is concentrated rather than general: São Paulo’s own transmission slope is an order of magnitude steeper than any other city’s over 2027–2031 (+7.4, $p < 0.001$) and flattens to +0.41 and insignificance by the tail window, which under the fixed-effects weighting described in Section 5.4.2 is by itself enough to account for the distance between the windows. The exercise therefore establishes something weaker than invariance, but sufficient for the use made of it here: the channel is positive and significant on every window tried, and the tail-window figure carried as the headline is the most conservative of the three. The mean level effect moves from +0.0002 to −0.0002 and back to approximately zero, but it is indistinguishable from zero in all three, which is the aggregate null of Section 5.1 restated on different data.

5.4.2. Slope heterogeneity and the pooling restriction

Equation 3 imposes a single slope β on all cities, allowing them to differ only in level through μ_i . That restriction is testable, and it fails. An F -test of the fixed-effects specification against one that interacts ΔM with the full set of city dummies rejects the common slope in both policy arms ($F(26, 478) = 2.93$ and $F(26, 744) = 4.69$, both $p < 10^{-5}$). Cities do not share a transmission coefficient; they share its sign in most cases, with a positive city-specific slope in 67 per cent of cities in the interest-only arm and 74 per cent in the improvement-active arm.

This matters for how the pooled estimate should be read. A fixed-effects estimator applied to heterogeneous slopes returns a weighted average of them, with weights proportional to how much ΔM happened to vary within each city, a quantity driven as much by stochastic divergence between seeds as by policy response. Those weights are concentrated: the three highest-weighted cities carry about half the estimate in either arm, and Porto Velho alone carries 34 and 30 per cent. The mean-group estimator, which averages the twenty-seven city-specific slopes with equal weight, gives +0.53 ($p = 0.051$) in the interest-only arm and +0.47 ($p = 0.017$) in the improvement-active arm, larger than the pooled coefficients of +0.31 and +0.37, and of the same sign.

The affordability channel is therefore positive under both estimators, and the typical city’s response is if anything larger than the pooled figure suggests. What the two estimators do not agree on is magnitude, and we report the pooled coefficient throughout as the more conservative of the two while noting that it is not the average city’s slope.

5.4.3. Sensitivity analysis

Five parameters that most directly condition the results above were varied one at a time over three levels straddling their defaults, at ten replications per level (150 simulations in all). Three were varied on Goiânia under the baseline configuration, and the two that act only through the improvement programme on Belém, where that programme operates at scale. Across the nine Goiânia cells the Gini coefficient stays within 0.442–0.448, a band no wider than the typical cell’s own seed-to-seed standard deviation, and unemployment, house vacancy, zero consumption and median affordability are similarly stable; only `NEIGHBORHOOD_EFFECT` moves any outcome monotonically beyond the seed noise, and it does so in the direction the affordability mechanism of Section 5.1 predicts. In Belém the improvement programme

responds to both of its own parameters, chiefly by clearing the initial stock of substandard dwellings faster or slower, and none of that response reaches the macro outcomes, where no end-to-end move clears the seed-to-seed variation within a cell. Appendix Appendix A.1 sets out the design, reports the full cell means (Table A.8), and states what the exercise does and does not establish.

6. Discussion

6.1. Limitations

Six limitations qualify these findings. The labour market produces unemployment below empirical levels (about 2.7 per cent against 4–15 per cent for Brazilian capitals) and housing production correspondingly above them (8–13 units per thousand inhabitants against 2–4); because lower unemployment compresses the lower tail of the income distribution, we read the reported magnitudes as conservative. The aggregate effects are small (mean $\Delta Y = +0.0002$, standard deviation 0.011) and individually significant in only six of the twenty-seven cities, so the affordability–inequality link holds as a conditional mechanism while the cross-city ordering is not identified by these data (Section 5.3). A common parameter set across cities buys comparability at the cost of city-specific calibration. The pooled mean ΔY has not settled onto a stable level by the averaging window, and $\hat{\beta}$ itself is about a third larger on earlier windows than on the tail (Section 5.4.1), so the headline magnitude should be read as the conservative end of a range rather than as a window-independent constant. The model runs in real terms with near-zero inflation while the regulated credit rates are specified in nominal terms, which may overstate the real cost of credit. And the improvement policy reaches broader affordability and inequality only through the model’s dynamic market-adjustment channels (Section 6.4), which are difficult to validate empirically.

6.2. Interpreting the null: mechanisms over averages

The average policy effect on inequality is statistically indistinguishable from zero. Read as a verdict that housing policy does not move inequality, this result would undercut the paper’s contribution; we instead read it as evidence that the distributional sign and magnitude of housing policy are contingent on how it reshapes affordability, rather than on whether a policy exists at all.

Two results anchor this framing. The same policy configurations that produce a zero average effect also produce a positive and significant $\hat{\beta}$ on the affordability channel ($p < 0.001$ at every geographic level, Table 3); the aggregate null is a composition effect, since the sample mixes configurations that raise the rental burden with configurations that ease it, each moving inequality in the opposite direction, so the average cancels out. The wealth channel compounds the cancellation rather than relieving it: it carries the opposite sign, so a configuration that raises both house prices and rents pushes inequality in two directions at once, and the net level effect is the residual of two transmissions rather than the absence of any.

The contribution is accordingly diagnostic rather than a verdict on whether to intervene: it identifies which margins of policy design determine the distributional outcome. A programme that expands housing access while leaving the rental-to-income burden unchanged, or worsening it, should not be expected to reduce inequality; one calibrated to ease that burden should. What the decomposition of Section 5.2 adds is that this is a property of the affordability margin itself rather than of a particular instrument: both levers examined reach it at the same rate, so the design question is what a programme does to housing costs, not which instrument it uses. The paper’s central finding is therefore not an average treatment effect but a mechanism: whether housing policy narrows or widens inequality is determined by how it reshapes the rental-to-income burden, not by whether it is implemented or which instrument it uses.

6.3. Why the wealth channel compresses

The house-price mechanism was the paper’s original hypothesis, and it is confirmed in operation but with the sign reversed relative to the literature reviewed in Section 2: policy-induced increases in house prices are associated with *lower* inequality, significantly so at the metropolitan and core levels. The wealth channel is usually theorised as the mechanism by which housing policy entrenches advantage [11; 2; 26], yet here it does the opposite.

The proximate reason is how housing wealth enters the outcome variable. Each family’s permanent income is $t_0 + r w$, wages plus a return r on total wealth w (savings, bank deposits, and owned real estate at current market price, net of loan balances), and it is this permanent income, smoothed over a 24-month window, on which the Gini coefficient is computed. A revaluation of the housing stock therefore raises the wealth component of permanent income roughly in proportion to the value each family already holds. Whether

that compresses or widens the distribution depends entirely on how concentrated housing wealth is relative to wages: adding weight to a source that is less unequally distributed than labour income lowers the overall Gini, and adding weight to a more unequally distributed source raises it. The negative coefficient says that in this model simulated housing wealth is the less concentrated of the two.

That is a plausible property of this particular institutional environment rather than a general claim. MCMV runs continuously in every configuration, including the baseline, and transfers dwellings free of charge to households in the bottom 38th percentile; the improvement programme upgrades the quality, and therefore the market value, of dwellings held by the same segment. Both instruments put housing assets in the hands of low-wage families, and they do so continuously over a thirty-year horizon. A distribution of housing assets that policy has broadened at the bottom, set against a more concentrated distribution of wages, is exactly the configuration in which asset appreciation compresses measured inequality. We do not decompose the Gini by income source to confirm this directly; that decomposition is the natural next test.

Two qualifications bound the reading. The transmission is weighted by r , which tracks the real (inflation-adjusted) policy interest rate and averages close to 0.1 % per month, so the channel converts only a small fraction of a capital gain into current permanent income; a one-standard-deviation policy-induced price movement (6.4 model units) maps to a Gini change of about -0.0013 , against a total ΔY standard deviation of 0.011. And the coefficient is not significant in the periphery, where it takes the same sign and a comparable magnitude (-1.44×10^{-4}) but is estimated with a wider standard error; we read that as a precision difference rather than a sign difference, but the core-only significance is what the data support.

6.4. Why two very different levers reach the distribution equally

The decomposition in Section 5.2 finds the regulated credit rate and the housing improvement programme transmitting to inequality through the affordability channel at indistinguishable rates ($\hat{\beta}_{\text{interest}} = 0.31$ vs. $\hat{\beta}_{\text{improvement}} = 0.37$, $p = 0.70$ on the difference). This is not the result the instruments' designs would suggest, and the asymmetry between them is worth setting out before explaining why it does not show up in the coefficient.

The two levers reach very different populations, continuously versus discretely. The regulated credit rate enters the purchasing-power screen for

every prospective buyer, every month, for the full simulation (Section 4.1). The housing improvement programme, by contrast, is re-applied monthly but to a narrow and shrinking pool: only households below the 38th income percentile whose dwelling sits exactly at the inadequate quality tier are eligible, the number upgraded each month is capped by the policy’s budget and by the construction capacity available to do the work, and once a house is upgraded it permanently exits the eligible pool. Of those three limits it is the eligible pool that binds in practice: the sensitivity exercise of Appendix Appendix A.1 finds the budget binding only while the initial stock of substandard dwellings is being cleared, and construction capacity never binding at all. On breadth of direct contact alone, the credit rate should dominate.

It does not, and the reason is that the improvement programme’s direct beneficiaries are not the whole of its footprint. The upgrade is contracted from a local construction firm and paid at the price the model would charge to build the same quality increment (Section 4), so each upgrade is simultaneously a quality transfer to one household and a demand injection into the construction sector, competing for the same capacity as new dwellings. That second effect propagates through wages, employment and the supply of new housing to families the programme never touches, and it reaches the rental market (where the affordability ratio is measured) through the same price and vacancy dynamics that the credit lever operates on. A narrow instrument with a broad fiscal footprint and a broad instrument with a narrow fiscal footprint can accordingly arrive at comparable coefficients.

This is consistent with how the model absorbs shocks more generally: PolicySpace3 does not clear its markets analytically, but through local, constructive rules: house prices respond to vacancy duration and negotiation, rents are discounted when unaffordable, and credit is rationed rather than perfectly elastic. The size of a policy-induced effect therefore depends on how far a lever’s spending travels through those adjustment rules, not on how many households it names as beneficiaries.

6.5. *Future work*

Density as a covariate. Section 4.4 reports the correlation between city population and mean ΔY graphically, without a formal regression, to keep the presentation focused on the affordability mechanism of Equation 3. A natural extension is to bring urban density (population per unit area) into the mechanism regression directly, interacted with the affordability channel,

to test whether the affordability–inequality relationship itself is conditioned by density net of city fixed effects. We leave this for future work because a credible density measure requires a defensible spatial unit (built-up area versus municipal boundary) that was outside this paper’s scope to construct and validate.

Per-series stabilisation / structural-break mapping. Section 5.4.1 establishes that $\hat{\beta}$ does not depend on where the averaging window is placed, but it says nothing about when, or whether, any individual trajectory settles. A natural extension would determine, for each (city, policy configuration, seed) series, the month at which the model stabilises, either in level or via detected structural breaks, and study how that stabilisation time varies across seeds, cities, and policy configurations. This is a substantially larger undertaking, requiring several hundred individual tests with attendant multiple-testing concerns, and we defer it as a self-contained follow-up.

7. Conclusions

The central contribution of this paper is the identification of the two channels through which housing policy interventions reach income inequality in Brazilian metropolitan areas, and of their opposite signs. Using an agent-based model that provides exact counterfactual comparisons across 1,596 simulations spanning twenty-seven cities, we show that the average policy effect on inequality is negligible, but that this aggregate null conceals two robust conditional mechanisms. Policy configurations that raise the rental burden increase inequality, while those that ease it reduce inequality; this relationship holds at all three geographic levels examined (full metropolitan area, capital municipality, and periphery) and is strongest in less dense peripheral areas, where lower-income households are more exposed to housing cost shocks. Policy configurations that raise house prices, by contrast, compress the distribution of permanent income, significantly so in the metropolitan area as a whole and in its urban core. The aggregate null is what remains after two transmissions of opposite sign have partly offset one another.

Two subsidiary findings sharpen the policy implications. First, the regulated credit rate and the housing improvement programme move affordability, and through it inequality, at statistically indistinguishable rates, so the instrument matters less than what it does to housing costs; a narrow programme whose spending reaches the construction sector can travel as far as a broad repricing of credit. Second, the distributional effects of housing

policy do not scale systematically with city size (metropolitan population is essentially uncorrelated with the magnitude of policy-induced Gini changes), suggesting that the affordability mechanism operates through local housing market conditions rather than through agglomeration intensity per se. We are, however, unable to say which cities respond most strongly: at the replication counts used here, only six of twenty-seven areas show individually significant effects, and the ordering of the rest is not identified.

These results carry a cautionary message for agglomeration-oriented urban strategies: the distributional reach of housing interventions depends less on the scale of the programme than on how it reshapes the cost of housing relative to income across the metropolitan income distribution. In this case, the housing-improvement programme’s own appreciation effect (Section 4) does not work against its stated aim as straightforwardly as one might assume, since asset appreciation is itself distributionally compressing in this setting, and the two levers arrive at the affordability channel on equal terms (Section 5.2). More broadly, policy-makers should treat housing policy as an intervention on a complex system [7], with significant indirect, non-trivial dynamics that must be addressed when designing interventions. We find that policies that improve housing quality or expand credit access without easing the rental burden may leave the distributional status quo unchanged, or even worsen it, which suggests the need for supportive policies aimed at stabilising or counteracting undesirable derived dynamics.

8. Acknowledgments

[This section is intentionally left blank for double-blind review.]

Appendix A. Supplementary figures and tables

Table A.5 re-estimates the Phillips-curve and Okun’s-law relationships of Section 3.3 (Table 1) separately within each of the six policy scenarios, as a stability check on the macroeconomic stylised facts targeted in calibration.

Table A.6 reports descriptive statistics (mean, standard deviation, minimum, median, maximum, and sample size) for the full set of simulated macroeconomic and housing-market indicators under the baseline configuration (medium interest rate, no housing improvement policy), pooled across the twenty-seven cities and their seeds over the evaluation window (2011–2021), consistent with the single baseline configuration reported per city in

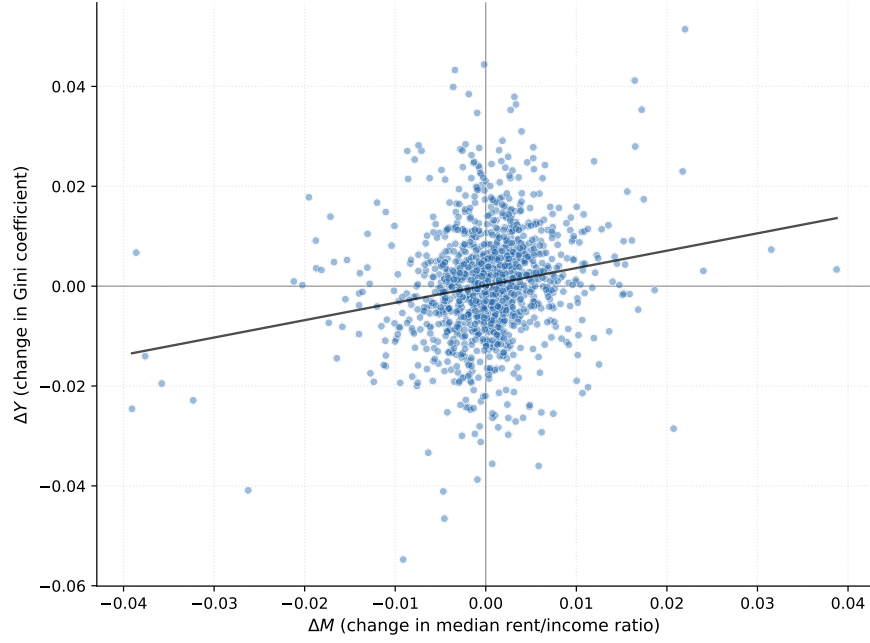


Figure A.3: Affordability channel: policy-induced change in the rental burden (ΔM , horizontal) against policy-induced change in the Gini coefficient (ΔY , vertical). Each point is one (metropolitan area, policy configuration, seed) observation ($N = 1,330$). The fitted line corresponds to Equation 3 with metropolitan-area fixed effects ($\hat{\beta} = +0.35$, $p < 0.001$). The width of the cloud relative to the slope is the $R^2 = 0.07$ reported in Table 3: the channel is a conditional relationship, not a description of most of the variance in ΔY .

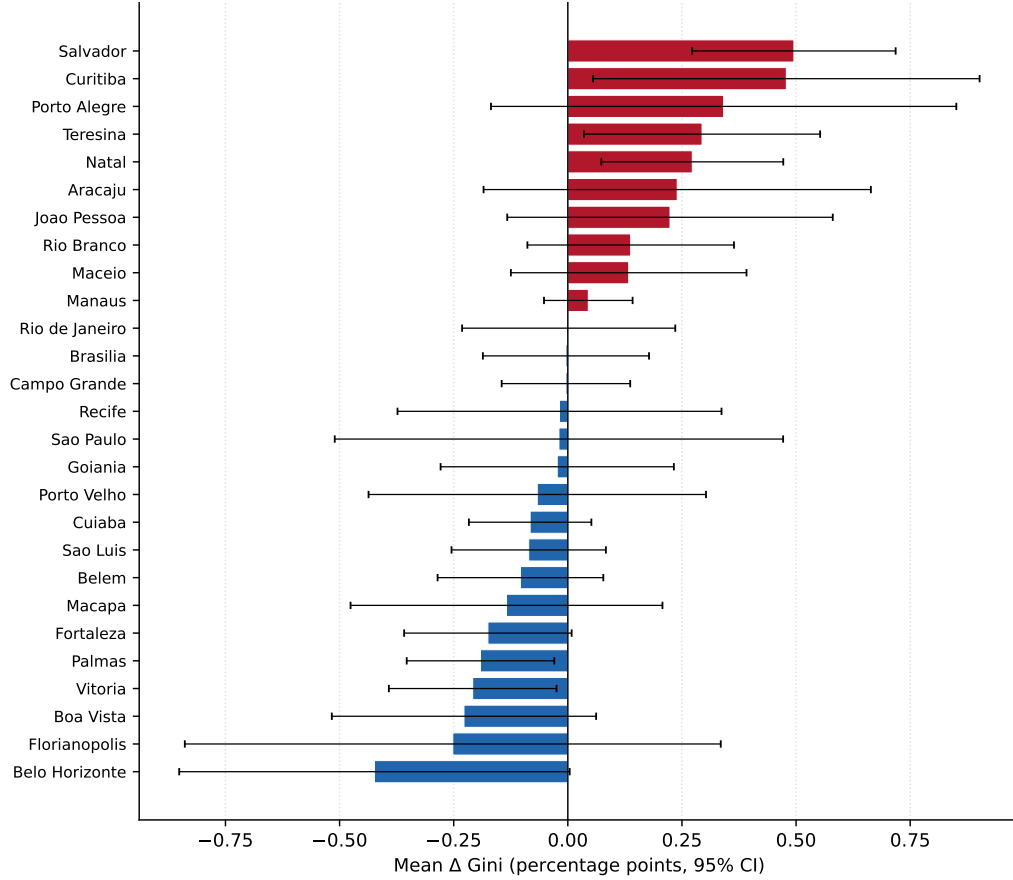


Figure A.4: Policy-induced change in the Gini coefficient (ΔY) for each of the twenty-seven metropolitan areas, averaged across all policy configurations and seeds ($N = 50$ per city, 30 for São Paulo). Bars are coloured by sign (blue: inequality reduction; red: increase). Error bars show 95% confidence intervals. Only six cities (Curitiba, Natal, Palmas, Salvador, Teresina and Vitória) have effects individually significant at the 5% level. The wide confidence intervals relative to the bar heights reflect seed-level stochastic variation, and are the reason the ordering of the bars is not interpreted (Section 5.3).

Table A.5: Phillips curve and Okun's law: stability across the six policy scenarios

Test	Scenario (rate / improvement policy)	$\hat{\beta}$	Std. error	p -value
Okun's law	High / Inactive	-0.182**	0.081	0.0330
Okun's law	High / Active	-0.162**	0.076	0.0440
Okun's law	Low / Inactive	-0.257***	0.083	0.0047
Okun's law	Low / Active	-0.209**	0.078	0.0130
Okun's law	Medium / Inactive	-0.190***	0.067	0.0088
Okun's law	Medium / Active	-0.219**	0.081	0.0123
Phillips curve	High / Inactive	-0.00131***	0.00013	< 0.001
Phillips curve	High / Active	-0.00118***	0.00015	< 0.001
Phillips curve	Low / Inactive	-0.00117***	0.00018	< 0.001
Phillips curve	Low / Active	-0.00128***	0.00017	< 0.001
Phillips curve	Medium / Inactive	-0.00116***	0.00018	< 0.001
Phillips curve	Medium / Active	-0.00126***	0.00015	< 0.001

Notes: same estimating equation as Table 1, with the slope estimated separately within each of the six policy scenarios (regulated credit rate $\times$ improvement policy status) rather than pooled, as a stability check on the sign and significance of both stylised facts.

Every one of the twelve scenario-level estimates is negative and significant at $p < 0.05$ (eight of the twelve at $p < 0.01$), indicating the macroeconomic regularities targeted in calibration hold within each policy configuration individually, not only in the pooled panel.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

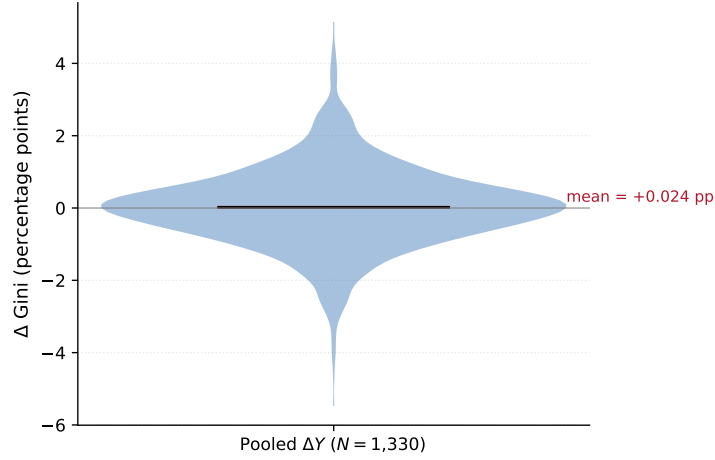


Figure A.5: Distribution of ΔY (policy-induced Gini change) across all 1,330 observations, pooled across cities, policy configurations, and seeds. The distribution is approximately symmetric around zero (mean = +0.024 pp, 48.5% negative), confirming that the aggregate policy effect is negligible. The standard deviation (1.11 pp) is more than an order of magnitude larger than the mean, indicating that within-city stochastic variation and cross-city heterogeneity dominate any average effect.

Table A.7 below. This is a descriptive summary of simulated output, not a comparison against target ranges.

Table A.7 reports simulated outcomes for each of the twenty-seven metropolitan areas individually under the baseline configuration, giving a descriptive view of the cross-city heterogeneity that underlies the pooled results in Section 5. It is not a calibration-fit table: several of the variables reported (unemployment, the Gini coefficient, vacancy, and the share of families with zero consumption) are the same stylised facts used to select the common parameter set described in Section 3.3, but no target ranges are shown here, only the simulated values themselves.

Appendix A.1. Sensitivity analysis

We test the five parameters that most directly condition the results above, chosen on theoretical grounds and calibration history rather than by screening: `UPGRADE_COST` (the share of a work's market value the state pays), `MAX_RENT_TO_INCOME_RATIO` (the affordability gate through which the paper's central mechanism operates), `MELHORIAS_INCOME_QUANTILE` (the improvement programme's eligibility threshold), `NEIGHBORHOOD_EFFECT` (how

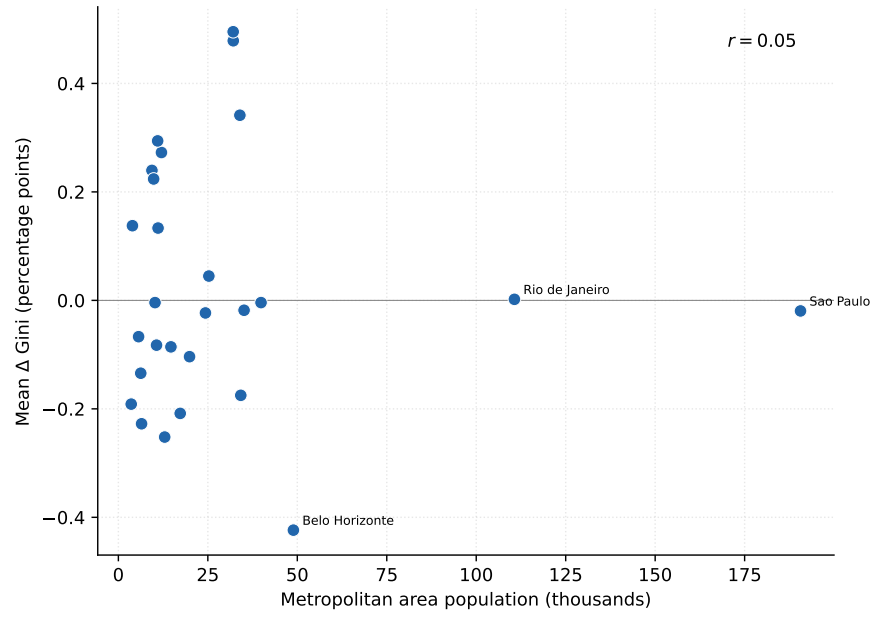


Figure A.6: Metropolitan area population (horizontal) against mean ΔY (vertical), one point per ACU, averaged across all configurations and seeds. The correlation between log population and mean ΔY is $r = +0.05$, offering no support for a systematic gradient in policy effectiveness with agglomeration intensity. Selected cities are labelled.

Table A.6: Descriptive statistics of simulated indicators under the baseline configuration (media interest rate, no housing improvement policy), evaluation window 2011–2021, pooled across 266 simulations in 27 cities.

Variable	Mean	Std	Min	Median	Max	N
Unemployment rate	0.0265	0.0083	0.0155	0.0250	0.1249	35,112
Gini coefficient	0.3564	0.0380	0.2859	0.3502	0.5452	35,112
Housing vacancy rate	0.0883	0.0125	0.0480	0.0892	0.1322	35,112
Zero consumption share	0.0445	0.0228	0.0025	0.0420	0.1428	35,112
GDP growth rate (monthly)	0.0108	0.0183	-0.1110	0.0110	0.1416	35,112
Inflation rate (monthly)	0.0002	0.0013	-0.0067	0.0002	0.0084	35,112
Quality of Life Index	0.7551	0.0399	0.6443	0.7543	0.8627	35,112
Avg. house price (model units)	177.8	22.6	131.9	175.6	309.2	35,112
Avg. rent (model units)	0.3282	0.0444	0.2334	0.3253	0.5150	35,112
Population	26187.5	35767.2	2295.0	13547.0	207537.0	35,112
Number of domiciles	10541.4	15076.1	755.0	5135.0	96474.0	35,112
Loan approval rate	0.6113	0.2283	0.0000	0.6380	1.0000	35,112
Active loans	542.8	815.8	0.0	281.0	5806.0	35,112
Credit stock (model units)	19456.2	23937.5	0.0	11238.9	152362.2	35,112
CO ₂ emissions (model units)	8578.8	9145.0	1267.3	5323.3	65919.2	35,112
Share firms profit > 0	0.8148	0.0441	0.7009	0.8129	0.9730	35,112
Wage per worker (model units)	1.86	0.40	0.42	1.88	2.90	35,112

strongly location quality capitalises into prices and rents), and **HOUSING_FINANCIAL_WEIGHT** (the opportunity cost of buying versus renting). Each is varied over three levels straddling its default, the other four held fixed, under the baseline credit-rate scenario with the improvement policy active, at ten replications per level and the same tail window (2035–2039) used throughout: 150 simulations in all. The sweeps run on two cities, Goiânia (a representative mid-sized capital) and Belém (where the improvement programme operates at scale); the two improvement-only parameters are varied on Belém, since Goiânia has almost no eligible households. Table A.8 reports the cell means.

Within each city the sweeps share their default (middle) cell, run under independent seeds, which fixes the noise floor against which any parameter effect must be read: the within-cell standard deviation of the Gini coefficient is 0.008 in Goiânia and 0.005 in Belém, comparable to the spread across cell means. Against that floor the outcome levels are stable. Across the nine Goiânia cells the Gini coefficient stays within 0.442–0.448, no wider than a typical cell’s own seed noise, and unemployment, house vacancy, zero-consumption share and median affordability are similarly contained; the six

Table A.7: Per-city simulated outcomes, baseline scenario (media interest rate, no housing improvement policy), evaluation window 2011–2021, averaged over the city’s replications.

City	Pop (k)	Unemp	Gini	Vacancy	ZeroCon	QLI	LoanAppr
Sao Paulo	205	0.023	0.448	0.089	0.045	0.817	0.588
Rio de Janeiro	120	0.022	0.384	0.079	0.059	0.765	0.457
Belo Horizonte	49	0.022	0.432	0.093	0.042	0.780	0.646
Recife	37	0.022	0.369	0.086	0.045	0.725	0.447
Brasilia	37	0.031	0.379	0.093	0.069	0.735	0.633
Porto Alegre	37	0.024	0.430	0.093	0.095	0.793	0.755
Salvador	35	0.023	0.360	0.089	0.046	0.739	0.452
Fortaleza	34	0.021	0.363	0.081	0.050	0.719	0.633
Curitiba	32	0.025	0.370	0.092	0.046	0.768	0.790
Goiania	22	0.026	0.339	0.092	0.041	0.763	0.964
Belem	21	0.027	0.337	0.074	0.036	0.732	0.317
Manaus	20	0.024	0.324	0.074	0.035	0.760	0.400
Vitoria	16	0.023	0.333	0.097	0.050	0.785	0.513
Sao Luis	14	0.025	0.340	0.091	0.039	0.740	0.803
Natal	12	0.025	0.342	0.086	0.038	0.735	0.698
Maceio	11	0.024	0.346	0.083	0.042	0.666	0.915
Teresina	10	0.028	0.343	0.089	0.037	0.733	0.670
Florianopolis	10	0.029	0.343	0.097	0.045	0.801	0.661
Joao Pessoa	9	0.033	0.339	0.088	0.054	0.731	0.834
Aracaju	9	0.024	0.353	0.087	0.043	0.708	0.816
Cuiaba	9	0.029	0.325	0.095	0.030	0.789	0.755
Campo Grande	9	0.031	0.327	0.087	0.039	0.808	0.996
Macapa	6	0.026	0.332	0.090	0.027	0.741	0.099
Porto Velho	5	0.027	0.372	0.103	0.046	0.730	0.407
Rio Branco	4	0.028	0.352	0.083	0.036	0.753	0.252
Boa Vista	4	0.028	0.330	0.087	0.027	0.779	0.398
Palmas	3	0.045	0.348	0.087	0.040	0.814	0.596
Mean	29	0.026	0.358	0.088	0.044	0.756	0.611

Belém cells are wider only because of seed noise, not parameter response.

Only one parameter moves an outcome monotonically beyond the noise. Raising `NEIGHBORHOOD_EFFECT` tightens the rental burden and pushes marginal families out of consumption (zero-consumption share, median affordability and rent default all rise end to end), the direction the mechanism of Section 5.1 predicts; the Gini itself does not follow, since the parameter moves affordability without moving the distribution of permanent income comparably. In Belém the improvement programme responds to its own two parameters only by clearing the initial stock of substandard dwellings faster or slower: `UPGRADE_COST` changes the early pace but the thirty-year totals converge, and `MELHORIAS_INCOME_QUANTILE` is slack over the lower half of its range. Which constraint binds changes over the run: the budget binds only early, while the backlog is being cleared, after which the eligible pool is the standing constraint; construction capacity never binds. None of this reaches the macro outcomes.

`HOUSING_FINANCIAL_WEIGHT` is inert over the lower part of its range, a property of running the model in real terms (Section 6.1) rather than an independent robustness result. This exercise tests whether simulated outcome levels stay in a plausible band under parameter perturbation, in two cities and one configuration; it does not re-estimate $\hat{\beta}$ in Equation 3 at each setting, which would require refitting the regression on cross-city, cross-configuration variation and is left for future work. Within that scope we find no evidence that the levels reported in this paper are an artefact of any single parameter choice examined.

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Table A.8: One-at-a-time sensitivity of tail-window (2035–2039) outcome levels

Level	Gini	Unemploy- ment	House vacancy	Zero consumption	Median affordability
<i>Panel A: Goiânia</i>					
MAX_RENT_TO_INCOME_RATIO					
0.2	0.4456 (0.0056)	0.0346 (0.0006)	0.1156 (0.0050)	0.0733 (0.0032)	0.0738 (0.0026)
0.3	0.4427 (0.0070)	0.0343 (0.0005)	0.1180 (0.0034)	0.0732 (0.0041)	0.0746 (0.0027)
0.4	0.4480 (0.0063)	0.0345 (0.0006)	0.1171 (0.0045)	0.0732 (0.0018)	0.0733 (0.0022)
NEIGHBORHOOD_EFFECT					
0.1	0.4424 (0.0093)	0.0343 (0.0009)	0.1156 (0.0026)	0.0713 (0.0038)	0.0724 (0.0028)
0.2	0.4418 (0.0094)	0.0346 (0.0010)	0.1137 (0.0036)	0.0738 (0.0037)	0.0744 (0.0024)
0.3	0.4469 (0.0074)	0.0344 (0.0006)	0.1143 (0.0039)	0.0760 (0.0036)	0.0774 (0.0029)
HOUSING_FINANCIAL_WEIGHT					
30	0.4440 (0.0072)	0.0346 (0.0009)	0.1167 (0.0038)	0.0742 (0.0030)	0.0733 (0.0018)
60	0.4440 (0.0072)	0.0346 (0.0009)	0.1167 (0.0038)	0.0742 (0.0030)	0.0733 (0.0018)
90	0.4484 (0.0080)	0.0348 (0.0007)	0.1174 (0.0041)	0.0738 (0.0026)	0.0741 (0.0029)
<i>Panel B: Belém</i>					
UPGRADE_COST					
0.5	0.3762 (0.0043)	0.0335 (0.0044)	0.1070 (0.0038)	0.0509 (0.0056)	0.0620 (0.0029)
1	0.3744 (0.0059)	0.0334 (0.0043)	0.1079 (0.0036)	0.0501 (0.0027)	0.0616 (0.0016)
1.5	0.3754 (0.0023)	0.0362 (0.0047)	0.1058 (0.0033)	0.0518 (0.0035)	0.0625 (0.0018)
MELHORIAS_INCOME_QUANTILE					
0.3	0.3769 (0.0037)	0.0396 (0.0101)	0.1099 (0.0043)	0.0516 (0.0052)	0.0613 (0.0029)
0.38	0.3762 (0.0042)	0.0390 (0.0104)	0.1100 (0.0043)	0.0506 (0.0058)	0.0609 (0.0029)
0.46	0.3798 (0.0041)	0.0468 (0.0134)	0.1069 (0.0051)	0.0554 (0.0066)	0.0633 (0.0030)

Notes: cell means over 2035–2039, ten replications per cell; within-cell seed-to-seed standard deviation in parentheses; defaults in bold. Panels are different cities, so levels are not comparable across them. The HOUSING_FINANCIAL_WEIGHT cells at 30 and 60 coincide by construction. Design and interpretation: see the text.

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